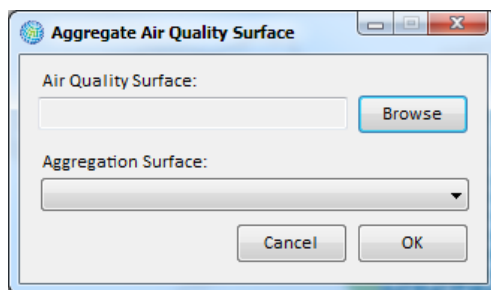


- **Monitor Data Conversion.** Convert monitor data files into BenMAP-CE format.
- **Neighbor File Creator.** This tool creates a text file (.txt) identifying "neighbor" monitors and associated interpolation weights for each grid cell in an air quality grid. Identifying the major contributors to air quality in a given cell can be helpful for understanding and troubleshooting BenMAP results, and for understanding how different geographic patterns of air quality changes can affect health benefits.
- **PopSim.** Simulate the cumulative effects of air pollution on different age groups over time (U.S. data only).
- **Options.** View and edit the general options for BenMAP-CE.
- **Compute Grid Crosswalks.** Remove or re-create crosswalks between grid definitions.

9.1 Air Quality Surface Aggregation

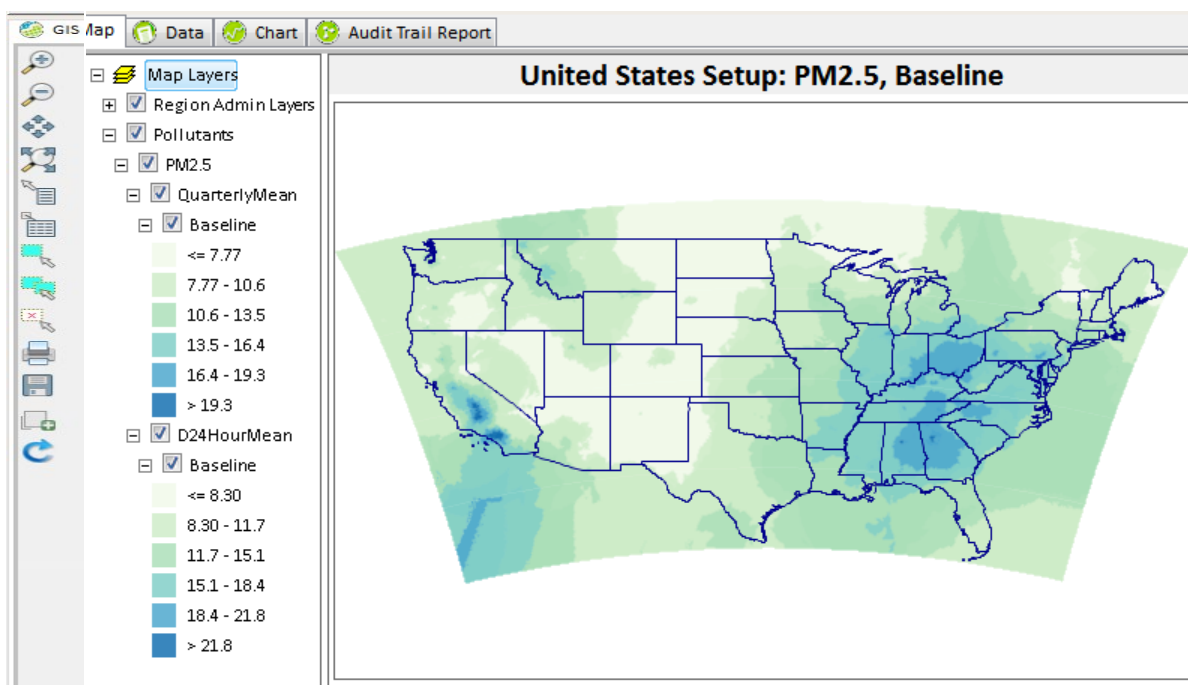
Using the **Aggregate Air Quality Surface** tool, you can change an air quality grid based on one grid definition to another (coarser) grid definition, using a simple spatially weighted average approach.

To start, choose *Aggregate Air Quality Surface* from the **Tools** drop-down menu. This will bring up the **Aggregate Air Quality Surface** window. Click the **Browse** button to find the air quality grid (.aqgx file) that you want to change and then use the **Aggregation Surface** drop-down list to select the new grid definition that you want to use. For example, you might want to aggregate a 12km model grid to the county-level. Click **OK** when done.

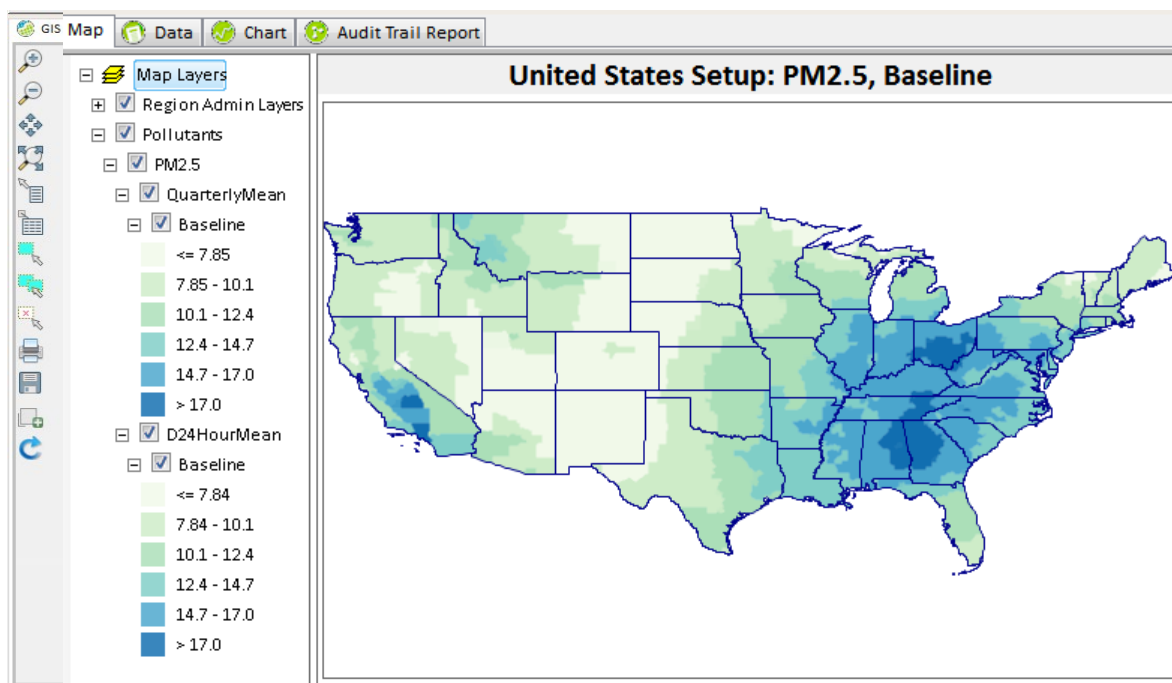


This will bring up the **Save Aggregated Air Quality Grid** window, where you specify the name of the newly aggregated air quality surface (.aqgx file) you are creating and its location. After the file is created, BenMAP-CE will return you to the main BenMAP-CE screen. You can then use the new file just as you would any other air quality surface.

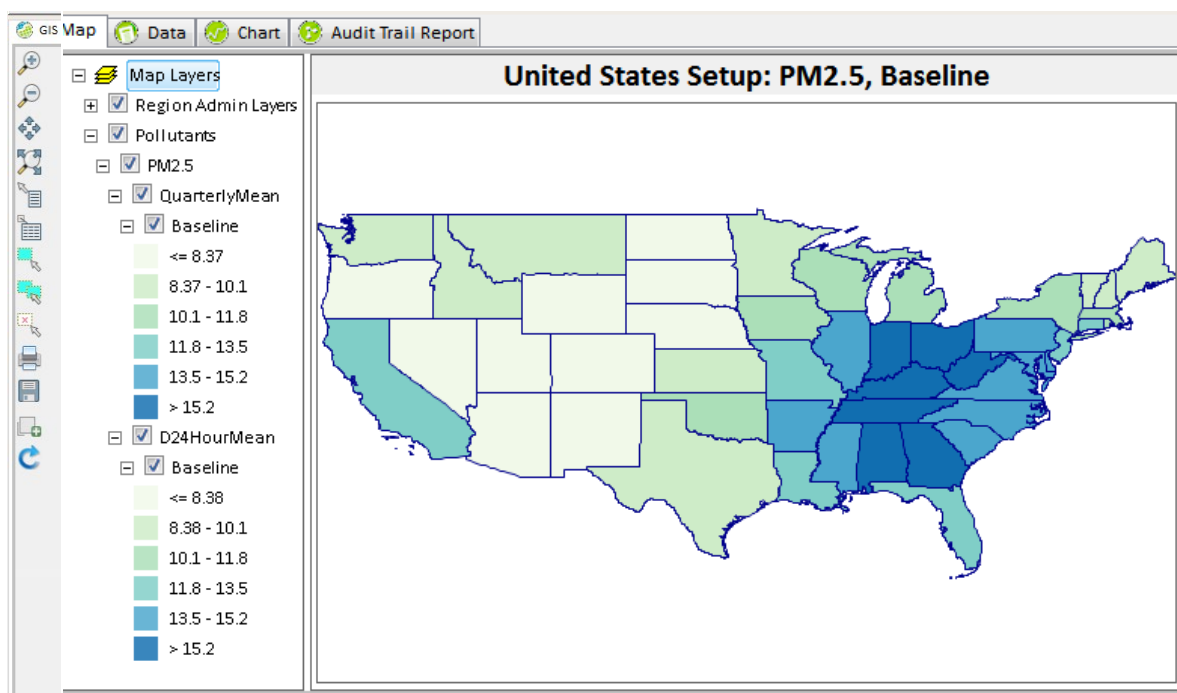
Below is an example of an air quality surface for PM_{2.5} created by the CMAQ model using a 12km grid. (The following images are cropped from the main BenMAP-CE window to focus on the GIS panel.)



Here is an example of the above 12km air quality surface aggregated to the county level.



And, here is an example of the 12km air quality surface aggregated to the state level.



Observe that there is more variation in the county-level file than the state-level file. This is expected, because BenMAP-CE is just using a simple spatially weighted average of the data.

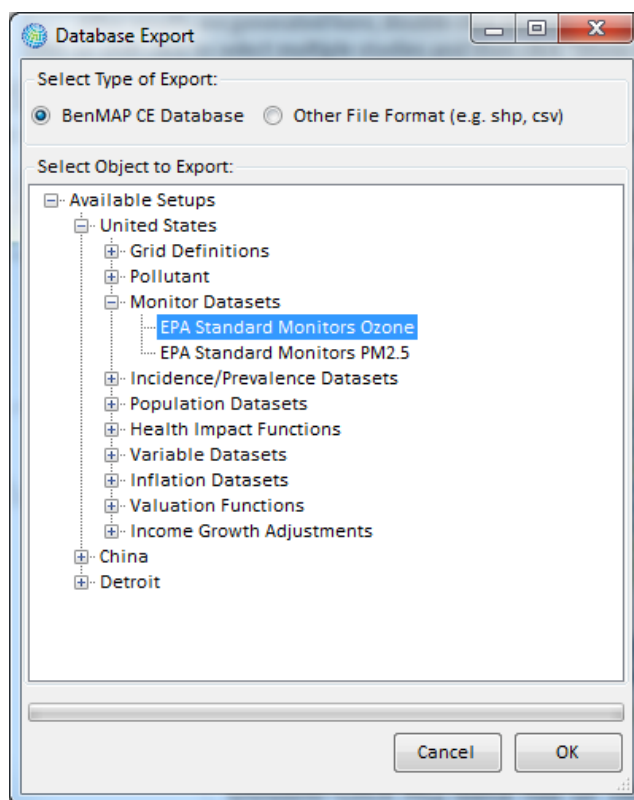
9.2 Database Export

BenMAP-CE allows you to export and import entire databases (all **Available Setups**), individual setups (e.g., *United States*, *China*), and parts of individual setups (e.g. all **Grid Definitions**, or individual **Health Impact Function** datasets). This functionality can be used to archive data, share data with other BenMAP-CE users, move databases between computers, as well as to view data in other applications like Excel or ArcMap. In particular, all of the steps involved in creating a setup can be done just once, after which the data can be exported and then imported on other computers. Version 1.5 also allows users to import databases exported from previous versions. This can be useful for transferring existing setups to the new version of BenMAP-CE without needing to re-create them.

To export part or all of an existing setup, go to the **Tools** menu, and choose the *Database Export* option.

This will bring up the **Database Export** window. Initially, all of the setups are listed in a tree menu, which is initially in a collapsed view. To expand any of the menu items, click on the '+' sign to the left of menu item. This will expand the tree menu to show additional listings for the expanded item. To collapse the tree menu, simply click on the '-' sign.

Choose a dataset to export by selecting it from the tree menu. The default Type of Export is BenMAP CE database. Press **OK**. In the screenshot below, we have chosen to export *EPA Standard Monitors O3* to a BenMap-CE database file.



This will bring up the **Save As** window. From here you may name the export file, select the export file type and choose the save location.